COMP262 - Phase 2: Machine Learning Sentiment Analysis

Generated on: 2026-04-19 18:42:10

Project Overview

This report was generated from previously saved model artifacts, figures, and evaluation outputs. It summarizes dataset preparation, sentiment classification model performance, and rating enhancement analysis.

Dataset Information

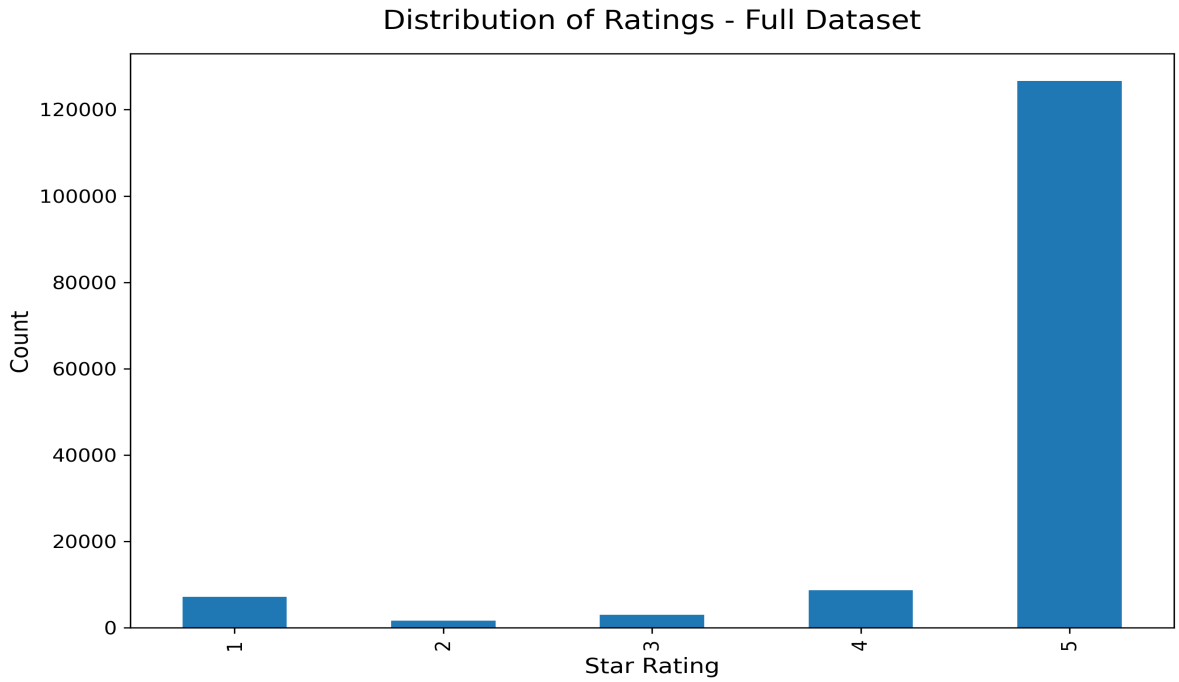
Metric	Value
Full dataset rows	147194
Balanced subset rows	8988
Training rows	6291
Test rows	2697
Subset class distribution	{'Negative': 2996, 'Positive': 2996, 'Neutral': 2996}
Training class distribution	{'Negative': 2097, 'Neutral': 2097, 'Positive': 2097}
Test class distribution	{'Negative': 899, 'Neutral': 899, 'Positive': 899}

Model Evaluation Summary

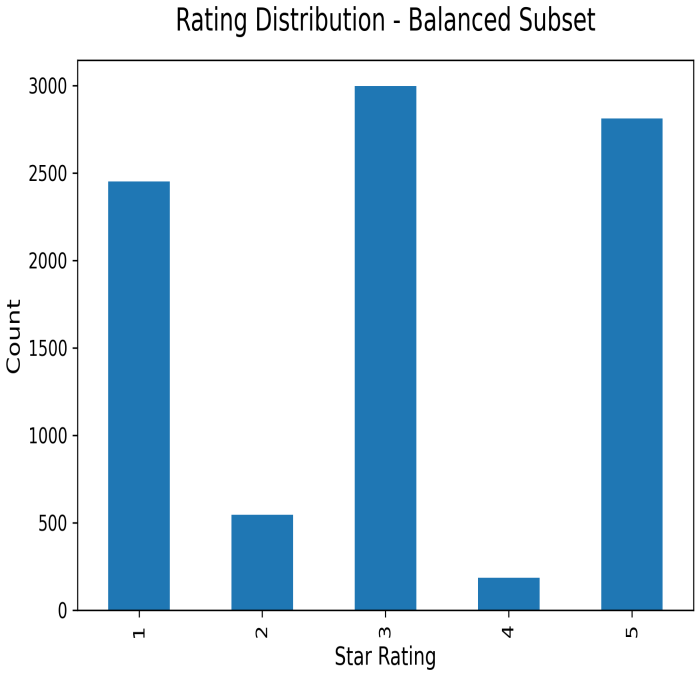
Model	Accuracy	Precision	Recall	F1
VADER	0.4320	0.4723	0.4320	0.3739
TextBlob	0.4846	0.4880	0.4846	0.4575
Naive Bayes	0.7078	0.7030	0.7078	0.7046
MLP	0.7097	0.7071	0.7097	0.7079

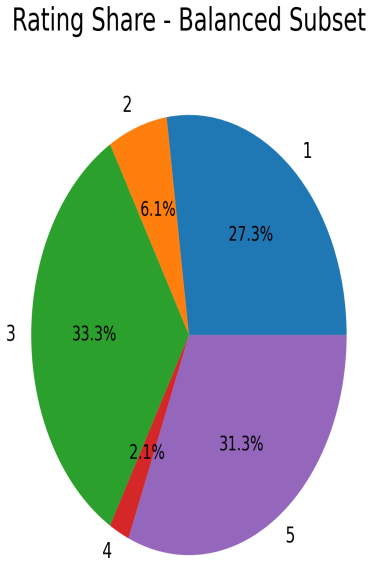
Figures

Distribution of Ratings - Full Dataset

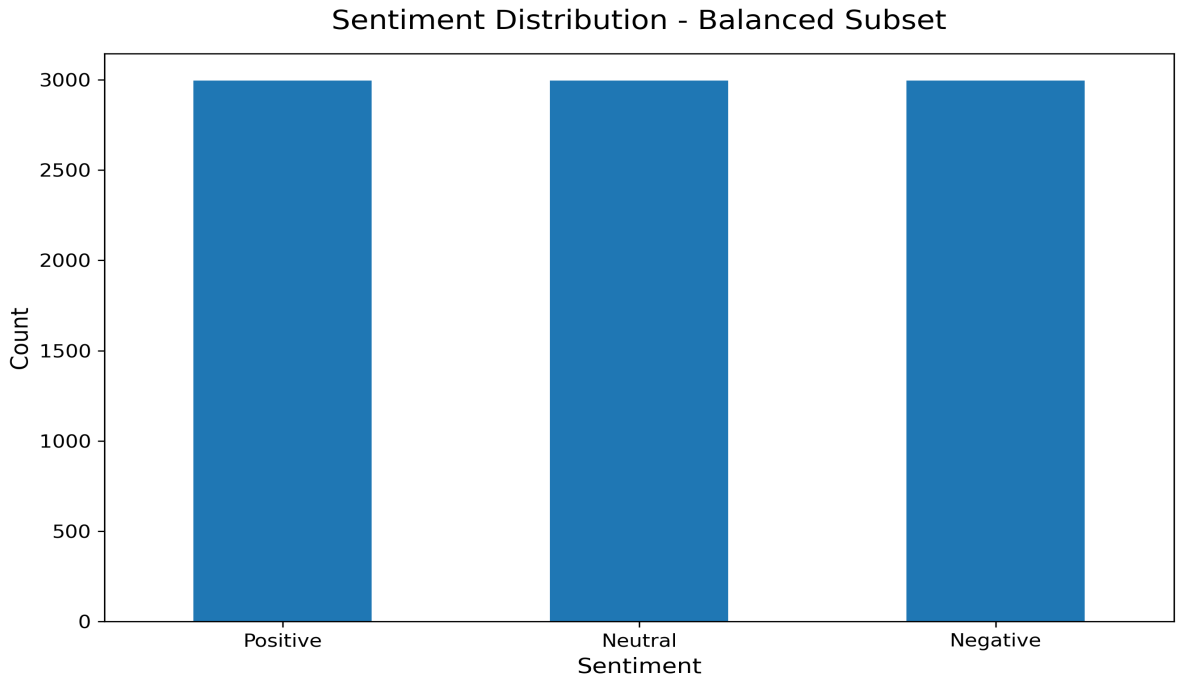


Rating Distribution - Balanced Subset

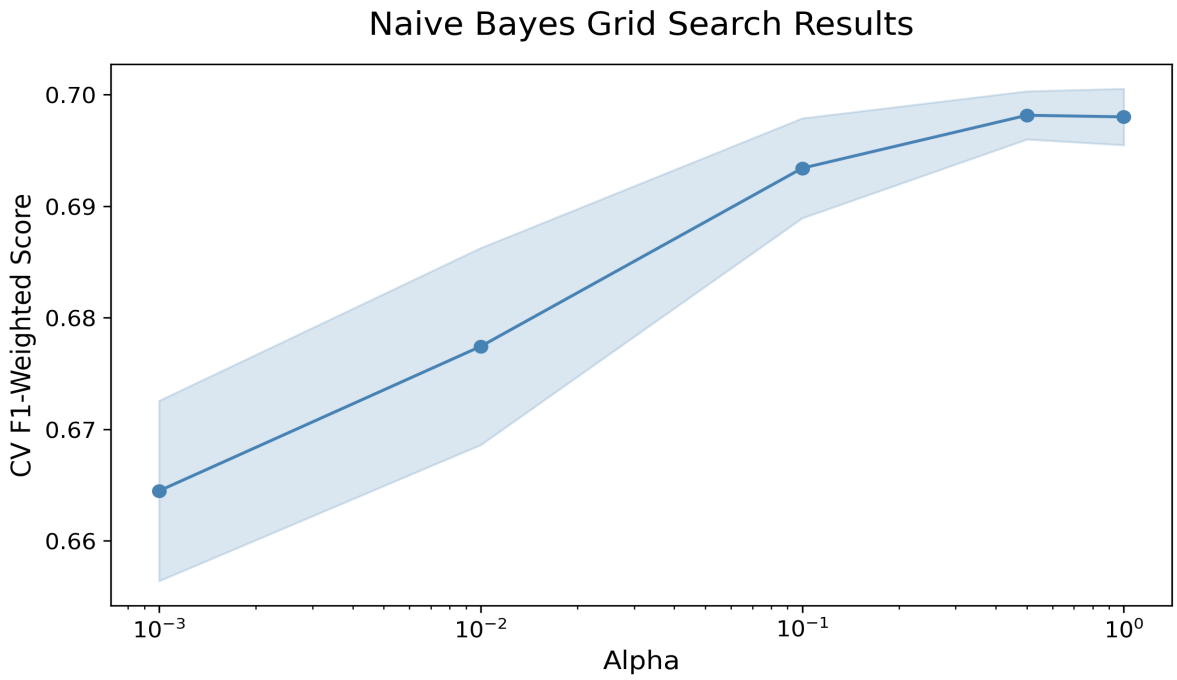




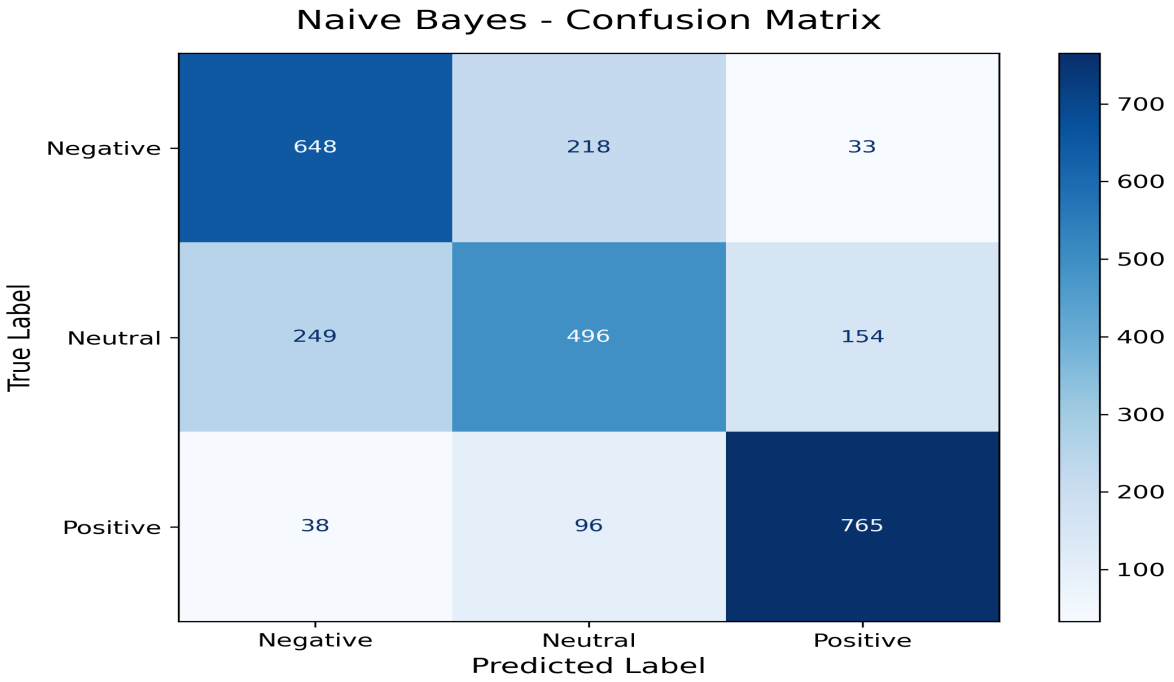
Sentiment Distribution - Balanced Subset



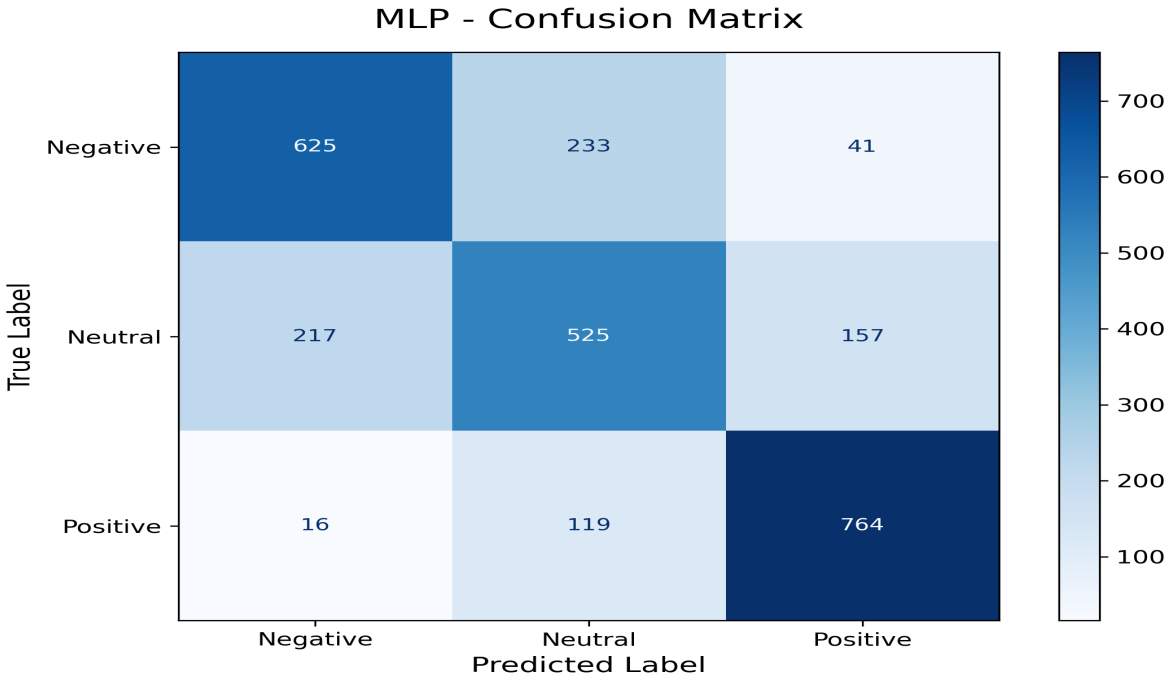
Naive Bayes Grid Search Results



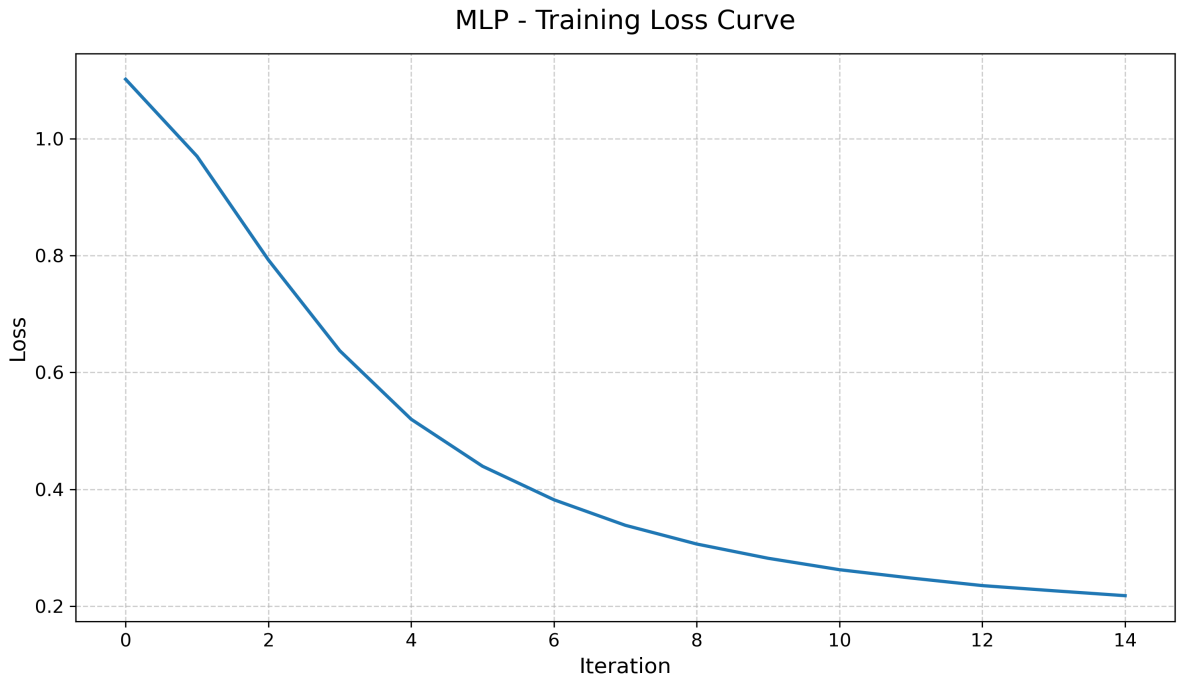
Naive Bayes Confusion Matrix



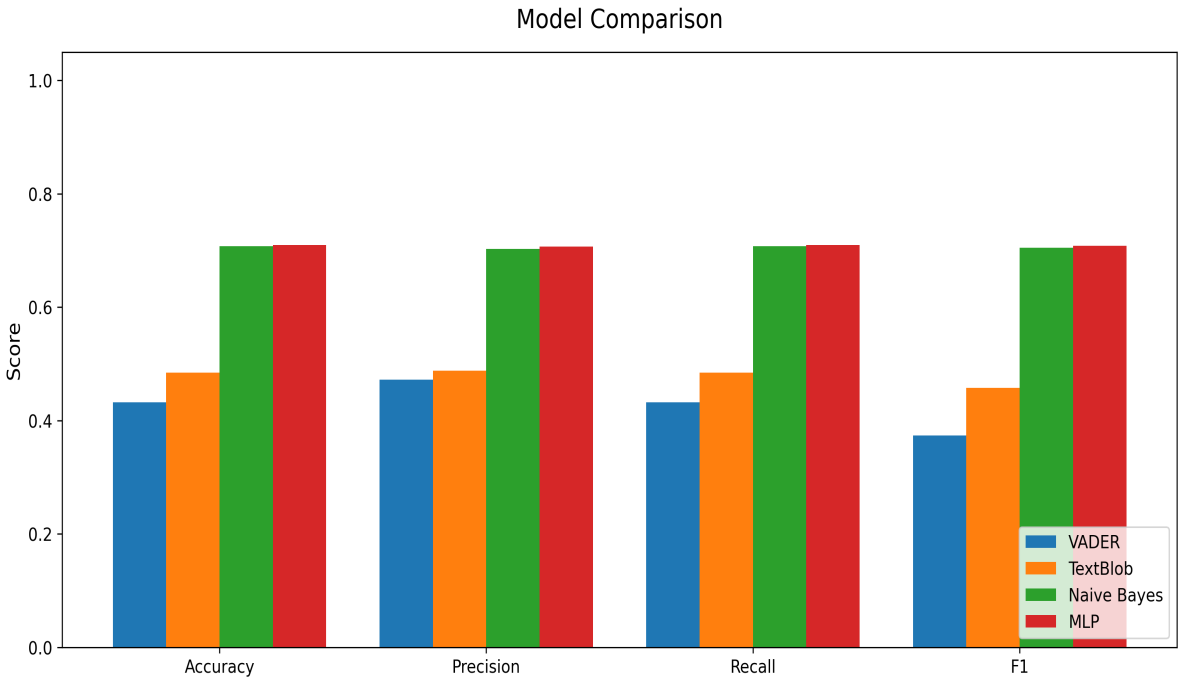
MLP Confusion Matrix



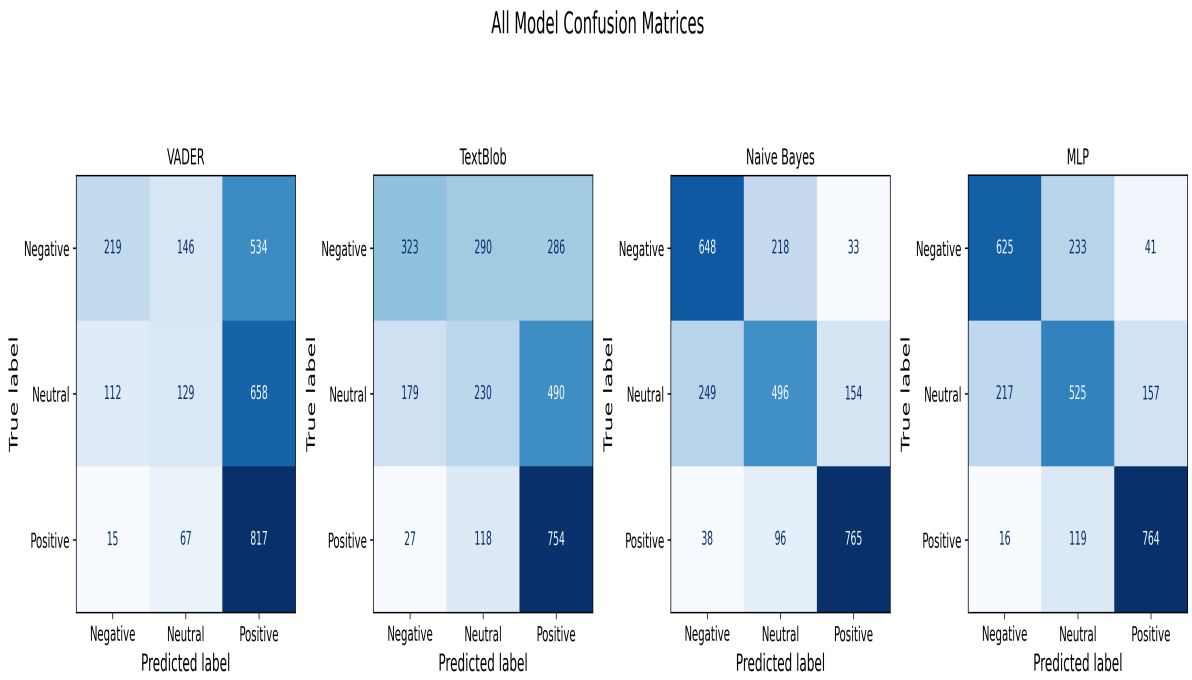
MLP Training Loss Curve



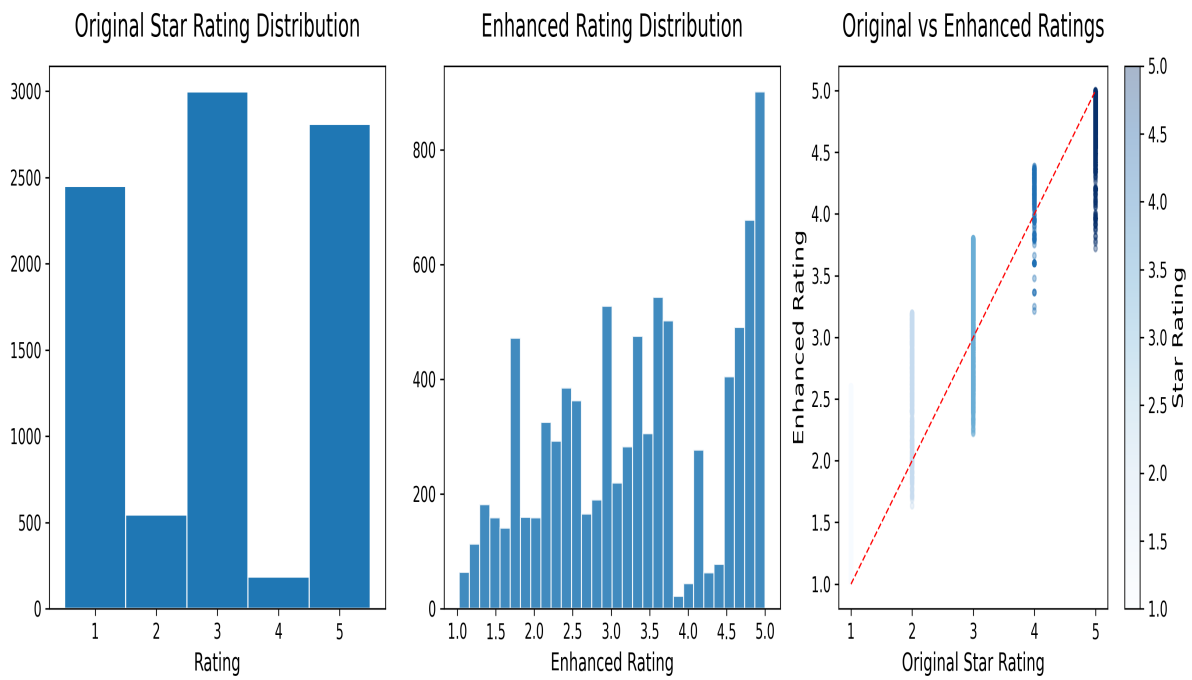
Model Comparison Bar Chart



All Model Confusion Matrices



Enhanced Rating Analysis



Classification Reports

VADER

```
precision    recall  f1-score   support

    Negative       0.63      0.24      0.35       899
     Neutral       0.38      0.14      0.21       899
    Positive       0.41      0.91      0.56       899

    accuracy                           0.43      2697
   macro avg       0.47      0.43      0.37      2697
weighted avg       0.47      0.43      0.37      2697
```

TextBlob

```
precision    recall  f1-score   support

    Negative       0.61      0.36      0.45       899
     Neutral       0.36      0.26      0.30       899
    Positive       0.49      0.84      0.62       899

    accuracy                           0.48      2697
   macro avg       0.49      0.48      0.46      2697
weighted avg       0.49      0.48      0.46      2697
```

Naive Bayes

```
precision    recall  f1-score   support

    Negative       0.69      0.72      0.71       899
     Neutral       0.61      0.55      0.58       899
    Positive       0.80      0.85      0.83       899

    accuracy                           0.71      2697
   macro avg       0.70      0.71      0.70      2697
weighted avg       0.70      0.71      0.70      2697
```

MLP

```
precision    recall  f1-score   support

    Negative       0.73      0.70      0.71       899
     Neutral       0.60      0.58      0.59       899
    Positive       0.79      0.85      0.82       899

    accuracy                           0.71      2697
   macro avg       0.71      0.71      0.71      2697
weighted avg       0.71      0.71      0.71      2697
```

Key Findings

- Best model by weighted F1: MLP (0.7079)

- Machine learning models outperform lexicon-based methods on the same test set.

- The balanced real subset provides a more reliable evaluation than earlier oversampled experiments.

- Neutral sentiment remains the hardest class, while Positive is generally the easiest to classify.